

Evaluation Part

Evaluation of two UML Diagrams

In this, we focusing on the paper named “New method for summative evaluation of UML class diagrams based on graph similarities”. This method for UML diagrams incorporates semantic meaning by analyzing relationships, context, and logical consistency beyond just syntactic and structural similarities. Semantic evaluation is performed by examining role-based connections, logical flow consistency, and validating terminology using domain-specific knowledge. The rubric-based evaluation integrates semantic correctness along with syntactic and structural checks. Each UML diagram is scored based on naming conventions, hierarchy validity, meaningful relationships, and logical consistency. By incorporating semantic meaning, this evaluation method ensures that UML diagrams are not only structurally correct but also logically meaningful.

Evaluation of Generated UML Diagram

The evaluation of the generated UML diagram involves assessing its correctness, completeness, and semantic alignment with the given scenario. This is achieved by comparing the textual representation of the UML diagram with the scenario description using NLP-based similarity measures. The TF-IDF Cosine Similarity method measures the lexical overlap between the two texts, providing an indication of structural similarity. Meanwhile, BERTScore evaluates the deep semantic alignment by leveraging contextual embeddings from a pre-trained BERT model. A higher similarity score suggests that the UML diagram accurately captures the key elements, relationships, and functional aspects described in the scenario.

BERTScore

BERTScore is an advanced similarity metric that evaluates text based on contextual meaning rather than just word overlap. It uses a pre-trained BERT model to generate deep embeddings for words and compares them based on precision, recall, and F1-score, capturing semantic relationships even if words are different but have similar meanings.

Cosine Similarity

Cosine similarity measures the textual similarity between two documents by converting them into TF-IDF vectors and calculating the cosine of the angle between them. It is a lexical-based metric that quantifies word overlap without considering meaning, making it effective for surface-level comparisons.

Link:

https://colab.research.google.com/drive/1C6eOKp6nc88i1r4b4xGxfO9ehMK_DSol?usp=sharing